

HDOC DAY-40

Title: String Handling Functions & Pass-by-Reference Operations using Pointers

1. Problem Statement

Write a menu-driven C program that reads a student's first name, middle name, and last name. Use separate user-defined functions for each operation. Use call by reference by passing pointers wherever a function modifies a string or returns multiple results.

Tasks to perform:

1. Combine first, middle, and last names into a full name using `strcpy()` and `strcat()`.
2. Find length of full name using `strlen()`.
3. Copy full name into another string using `strcpy()`.
4. Compare first and last names using `strcmp()`.
5. Display name in reverse order (Last, Middle, First) using `strcpy()` and `strcat()`.
6. Count vowels, consonants, digits, and spaces using `isalpha()`, `isdigit()`, `isspace()`, `tolower()`, and `strlen()`.
7. Search if a word occurs in the full name using `strstr()`.

8. Exit.

2. Header Files Used

- <stdio.h>: Input/output operations (printf, scanf).
- <string.h>: String manipulation functions (strcpy, strcat, strlen, strcmp, strstr).
- <ctype.h>: Character classification utilities (isalpha, isdigit, isspace, tolower).

3. Test Case Table

Test Case	Selected Task	Input	Expected Output	Status
1	Combine Names	Hardik, Kumar, Suhag	"Hardik Kumar Suhag"	Pass
2	Full Name Length	"Hardik Kumar Suhag"	18 characters	Pass
3	Copy Name	"Hardik Kumar Suhag"	Copied: "Hardik Kumar Suhag"	Pass

Test Case	Selected Task	Input	Expected Output	Status
4	Compare Names	Hardik vs Suhag	First and Last names are DIFFERENT	Pass
5	Reverse Order	Hardik, Kumar, Suhag	"Suhag Kumar Hardik"	Pass
6	Character Count	"Hardik Kumar Suhag"	Vowels: 6, Consonants: 10, Digits: 0, Spaces: 2	Pass
7	Word Search	Word: "Kumar"	Word WAS FOUND in full name	Pass
8	Exit	Option 8	Program terminates cleanly	Pass

4. Source Code Listing

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
```

```
void combineNames(const char firstName[], const char middleName[],
const char lastName[], char fullName[]) {
    strcpy(fullName, firstName);
    strcat(fullName, " ");
    strcat(fullName, middleName);
    strcat(fullName, " ");
    strcat(fullName, lastName);
}
```

```
void findLength(const char fullName[], size_t *length) {
    *length = strlen(fullName);
}
```

```
void copyName(const char source[], char destination[]) {
    strcpy(destination, source);
}
```

```
void compareNames(const char firstName[], const char lastName[], int
*comparisonResult) {
    *comparisonResult = strcmp(firstName, lastName);
}
```

```
void reverseNameOrder(const char firstName[], const char
middleName[], const char lastName[], char reverseName[]) {
    strcpy(reverseName, lastName);
    strcat(reverseName, " ");
    strcat(reverseName, middleName);
    strcat(reverseName, " ");
    strcat(reverseName, firstName);
}
```

```
void countCharacters(const char fullName[], int *vowels, int
*consonants, int *digits, int *spaces) {
    *vowels = 0; *consonants = 0; *digits = 0; *spaces = 0;

    for (size_t i = 0; i < strlen(fullName); i++) {
        char ch = fullName[i];
        if (isdigit((unsigned char)ch)) {
            (*digits)++;
        }
    }
}
```

```

    } else if (isspace((unsigned char)ch)) {
        (*spaces)++;
    } else if (isalpha((unsigned char)ch)) {
        char lower = tolower((unsigned char)ch);
        if (lower == 'a' || lower == 'e' || lower == 'i' || lower == 'o' || lower == 'u')
        {
            (*vowels)++;
        } else {
            (*consonants)++;
        }
    }
}
}
}

```

```

void searchWord(const char fullName[], const char word[], int
*searchResult) {
    *searchResult = (strstr(fullName, word) != NULL) ? 1 : 0;
}

```

```

int main(void) {
    char firstName[50], middleName[50], lastName[50];
    char fullName[150] = "", copiedName[150] = "", reverseName[150] = "";
    int choice;

    printf("Enter First Name: "); scanf("%s", firstName);
    printf("Enter Middle Name: "); scanf("%s", middleName);
    printf("Enter Last Name: "); scanf("%s", lastName);

    combineNames(firstName, middleName, lastName, fullName);

    do {
        printf("\n1. Combine Names\n2. Find Length\n3. Copy Name\n4.
Compare Names\n");
        printf("5. Reverse Order\n6. Count Characters\n7. Search Word\n8.
Exit\nChoice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                combineNames(firstName, middleName, lastName, fullName);

```

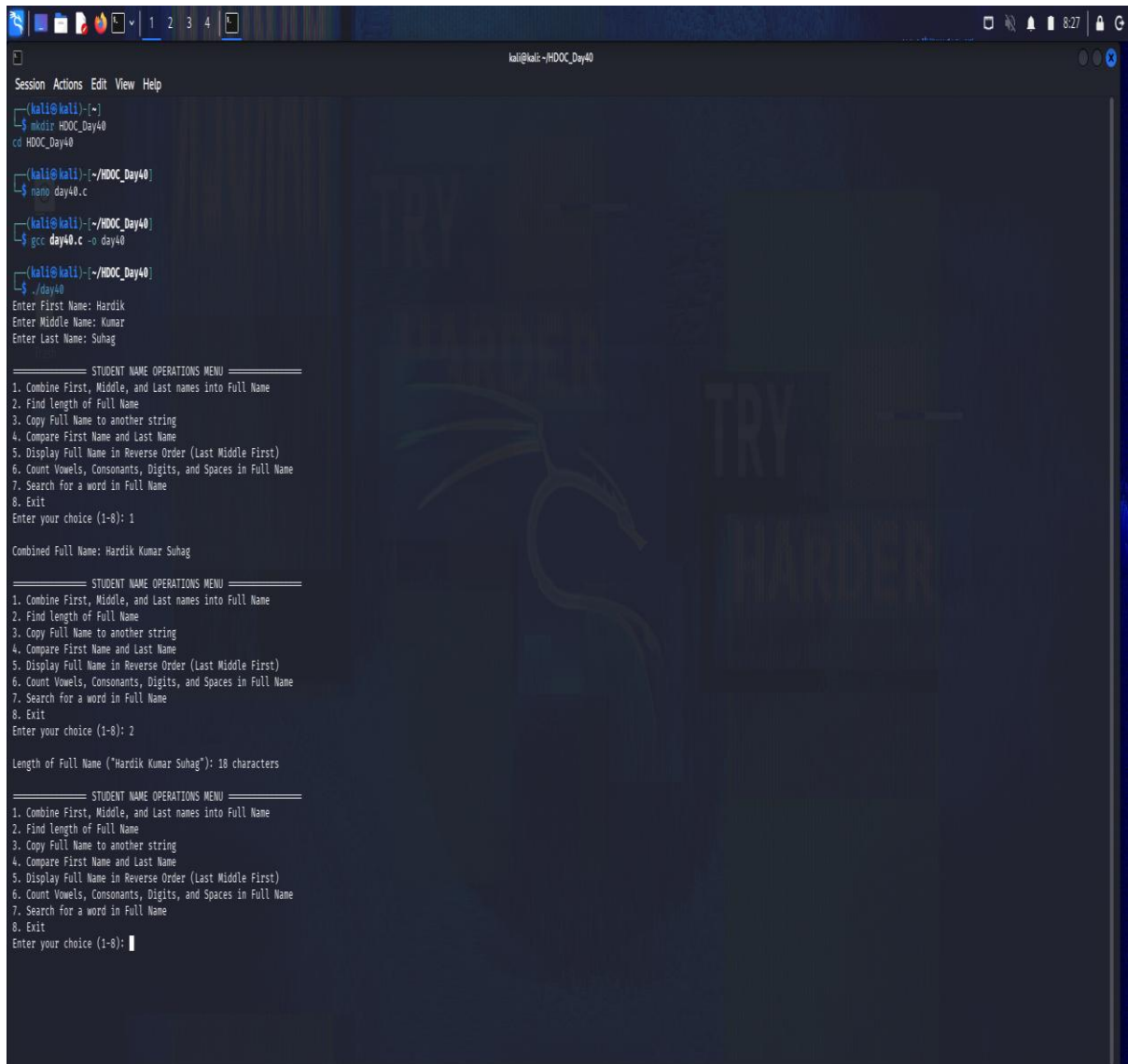
```

        printf("Full Name: %s\n", fullName);
        break;
    case 2: {
        size_t length = 0;
        findLength(fullName, &length);
        printf("Length: %zu\n", length);
        break;
    }
    case 3:
        copyName(fullName, copiedName);
        printf("Copied String: %s\n", copiedName);
        break;
    case 4: {
        int res = 0;
        compareNames(firstName, lastName, &res);
        printf("Comparison: %s\n", (res == 0) ? "EQUAL" : "DIFFERENT");
        break;
    }
    case 5:
        reverseNameOrder(firstName, middleName, lastName,
reverseName);
        printf("Reversed Order: %s\n", reverseName);
        break;
    case 6: {
        int v = 0, c = 0, d = 0, s = 0;
        countCharacters(fullName, &v, &c, &d, &s);
        printf("Vowels: %d, Consonants: %d, Digits: %d, Spaces: %d\n",
v, c, d, s);
        break;
    }
    case 7: {
        char word[50]; int res = 0;
        printf("Search word: "); scanf("%s", word);
        searchWord(fullName, word, &res);
        printf("Result: %s\n", (res == 1) ? "FOUND" : "NOT FOUND");
        break;
    }
    case 8: printf("Exiting...\n"); break;
    default: printf("Invalid choice!\n"); break;
}

```

```
} while (choice != 8);  
  
return 0;  
}
```

5. Execution Screenshots



The screenshot shows a terminal window titled 'kali@kali:~/HDOC_Day40'. The user navigates to the directory 'HDOC_Day40' and creates a file 'day40.c' using 'nano'. The file is then compiled with 'gcc day40.c -o day40'. The user runs the program './day40', which prompts for a first name ('Hardik'), middle name ('Kumar'), and last name ('Suhag'). The program then displays a menu titled 'STUDENT NAME OPERATIONS MENU' with 8 options. The user selects option 1, which outputs 'Combined Full Name: Hardik Kumar Suhag'. The user then selects option 2, which outputs 'Length of Full Name ("Hardik Kumar Suhag"): 18 characters'. The menu is displayed again, and the user selects option 2, which outputs 'Length of Full Name ("Hardik Kumar Suhag"): 18 characters'.

```
kali@kali:~/HDOC_Day40  
$ pwd  
/HDOC_Day40  
$ cd HDOC_Day40  
$ nano day40.c  
$ gcc day40.c -o day40  
$ ./day40  
Enter First Name: Hardik  
Enter Middle Name: Kumar  
Enter Last Name: Suhag  
  
===== STUDENT NAME OPERATIONS MENU =====  
1. Combine First, Middle, and Last names into Full Name  
2. Find Length of Full Name  
3. Copy Full Name to another string  
4. Compare First Name and Last Name  
5. Display Full Name in Reverse Order (Last Middle First)  
6. Count Vowels, Consonants, Digits, and Spaces in Full Name  
7. Search for a word in Full Name  
8. Exit  
Enter your choice (1-8): 1  
  
Combined Full Name: Hardik Kumar Suhag  
  
===== STUDENT NAME OPERATIONS MENU =====  
1. Combine First, Middle, and Last names into Full Name  
2. Find Length of Full Name  
3. Copy Full Name to another string  
4. Compare First Name and Last Name  
5. Display Full Name in Reverse Order (Last Middle First)  
6. Count Vowels, Consonants, Digits, and Spaces in Full Name  
7. Search for a word in Full Name  
8. Exit  
Enter your choice (1-8): 2  
  
Length of Full Name ("Hardik Kumar Suhag"): 18 characters  
  
===== STUDENT NAME OPERATIONS MENU =====  
1. Combine First, Middle, and Last names into Full Name  
2. Find Length of Full Name  
3. Copy Full Name to another string  
4. Compare First Name and Last Name  
5. Display Full Name in Reverse Order (Last Middle First)  
6. Count Vowels, Consonants, Digits, and Spaces in Full Name  
7. Search for a word in Full Name  
8. Exit  
Enter your choice (1-8):
```